

20260401 Updated Proxy EW(H α)

1. Updated calculation of $\text{EW}_{\text{proxy}}(\text{H}\alpha)$

Here we now compute $\text{EW}_{\text{proxy}}(\text{H}\alpha)$ from the observed H α line flux and the continuum-only cube `*_CONTcube.fits`, instead of using only the broadband R -band flux.

The observed H α flux map is taken from the `HA6562_FLUX` extension in either `*_gas_BIN_maps_extended.fits` or `*_gas_BIN_maps.fits`. The line-of-sight H α velocity map is taken from the corresponding `HA6562_VEL` extension in the same gas file.

For the continuum window, we follow the [MaNGA DAP](#) H α EW passband in vacuum,

$$\lambda_{\text{vac,low}} = 6557.6 \text{ \AA}, \quad \lambda_{\text{vac,high}} = 6571.6 \text{ \AA}, \quad (1)$$

with the H α Ritz wavelength

$$\lambda_{\text{vac,ref}} = 6564.608 \text{ \AA}. \quad (2)$$

Because the local continuum cube wavelength axis is in standard air wavelength, these values are converted from vacuum to air using the NIST ASD / Peck & Reeder (1972) refractive-index formula,

$$\lambda_{\text{air}} = \lambda_{\text{vac}}/n, \quad (3)$$

with

$$(n - 1) \times 10^8 = 8060.51 + \frac{2480990}{132.274 - \sigma^2} + \frac{17455.7}{39.32957 - \sigma^2}, \quad (4)$$

where σ is the vacuum wavenumber in inverse microns. The Ritz wavelength in vacuum is related to wavenumber through

$$\lambda_{\text{vac}} [\text{nm}] = \frac{10^7}{\sigma [\text{cm}^{-1}]}, \quad (5)$$

which gives the rest-frame air H α reference wavelength and EW window used by the code.

Therefore, we have

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HalpHa window (vacuum, rest) : 6557.600 - 6571.600 A
HalpHa window (air, rest)    : 6555.789 - 6569.785 A
HalpHa lambda (air, rest)    : 6562.795 A
HalpHa Delta lambda (air)   : 13.996 A

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The **key update** in the current implementation is that the continuum window is no longer shifted using only a single galaxy redshift. Instead, the galaxy redshift z_{gal} from `new_redshifts` is first converted to a systemic relativistic velocity,

$$\beta_{\text{gal}} = \frac{(1 + z_{\text{gal}})^2 - 1}{(1 + z_{\text{gal}})^2 + 1}, \quad v_{\text{gal}} = c \beta_{\text{gal}}. \quad (6)$$

Then, for each spaxel (x, y) , the observed H α velocity is added on top of the systemic velocity,

$$v_{\text{spaxel}}(x, y) = v_{\text{gal}} + v_{\text{H}\alpha}(x, y). \quad (7)$$

This total velocity is converted back to a relativistic spaxel redshift,

$$\beta_{\text{spaxel}}(x, y) = \frac{v_{\text{spaxel}}(x, y)}{c}, \quad z_{\text{spaxel}}(x, y) = \sqrt{\frac{1 + \beta_{\text{spaxel}}(x, y)}{1 - \beta_{\text{spaxel}}(x, y)}} - 1. \quad (8)$$

Using this spaxel-dependent redshift, the H α EW window is re-adjusted separately for every spaxel in the observed frame:

$$\lambda_{\text{obs},1}(x, y) = \lambda_{\text{rest},1} [1 + z_{\text{spaxel}}(x, y)], \quad (9)$$

$$\lambda_{\text{obs},2}(x, y) = \lambda_{\text{rest},2} [1 + z_{\text{spaxel}}(x, y)]. \quad (10)$$

At the same time, the observed continuum flux density is converted to rest-frame flux density per unit wavelength as

$$f_{\lambda,\text{rest}}(x, y) = [1 + z_{\text{spaxel}}(x, y)] f_{\lambda,\text{obs}}(x, y). \quad (11)$$

The mean continuum flux density in the H α EW window is therefore measured separately for each spaxel from all continuum-cube planes whose observed wavelengths fall inside that spaxel's shifted H α bandpass:

$$\bar{f}_{\lambda,\text{cont}}(x, y) = \frac{1}{\Delta\lambda} \int_{\lambda_1}^{\lambda_2} f_{\lambda,\text{rest}}(x, y; \lambda) d\lambda. \quad (12)$$

In practice, this is evaluated as the sampled mean of the finite continuum-cube values inside the spaxel-dependent H α window.

The new proxy equivalent width is therefore

$$\text{EW}_{\text{proxy}}(\text{H}\alpha)(x, y) = \frac{F_{\text{H}\alpha}(x, y)}{\bar{f}_{\lambda, \text{cont}}(x, y)}. \quad (13)$$

For comparison, the old broadband-based pseudo EW is still preserved in a separate HDU. In that legacy approach, the observed R -band magnitude

`MAGNITUDE_R_UNCORRECTED` from `*_SPATIAL_BINNING_maps_extended.fits` is first converted to nanomaggies,

$$R_{\text{nanomaggy}} = 10^{(22.5 - m_{R, \text{obs}})/2.5}, \quad (14)$$

then to flux density per unit frequency,

$$f_{\nu} = R_{\text{nanomaggy}} \times 3.631 \times 10^{-29} \quad [\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}], \quad (15)$$

and then to a single-wavelength continuum estimate at $6562.8 \text{ \AA}$,

$$f_{\lambda, \text{legacy}} = f_{\nu} c / \lambda^2. \quad (16)$$

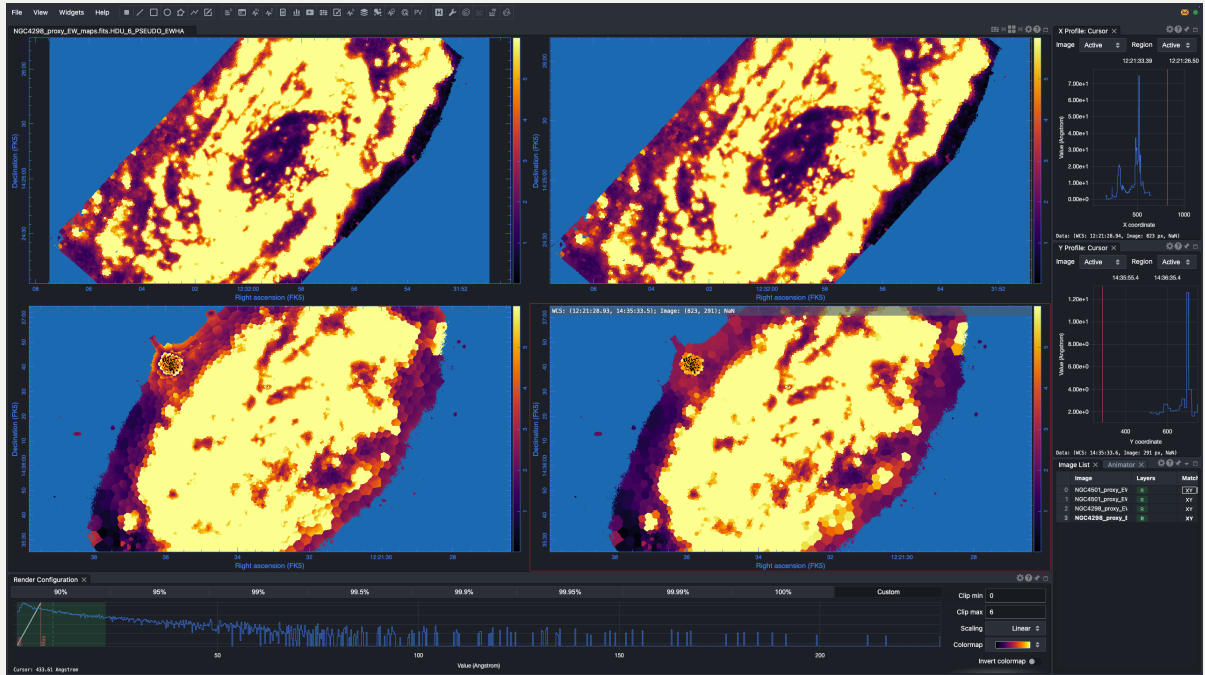
The old pseudo EW is therefore

$$\text{EW}_{\text{pseudo}}(\text{H}\alpha) = \frac{F_{\text{H}\alpha}}{f_{\lambda, \text{legacy}}}. \quad (17)$$

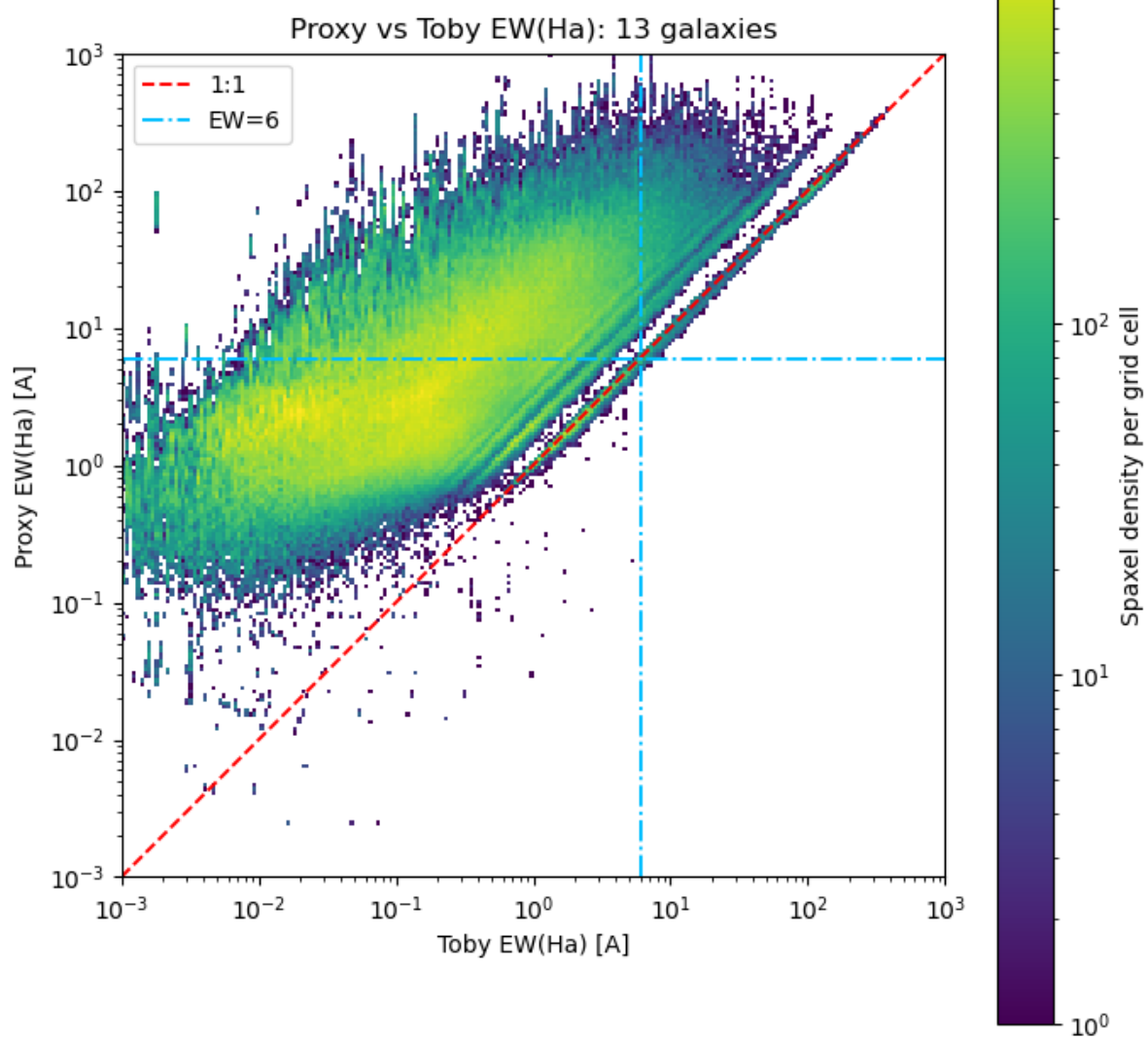
Note that $\text{EW}_{\text{proxy}}(\text{H}\alpha)$ is now based on the continuum-only spectrum measured directly across the $\text{H}\alpha$ EW window, with the observed window re-centered separately for each spaxel using `HA6562_VEL`, whereas $\text{EW}_{\text{pseudo}}(\text{H}\alpha)$ is the older single-band approximation from the broadband R flux.

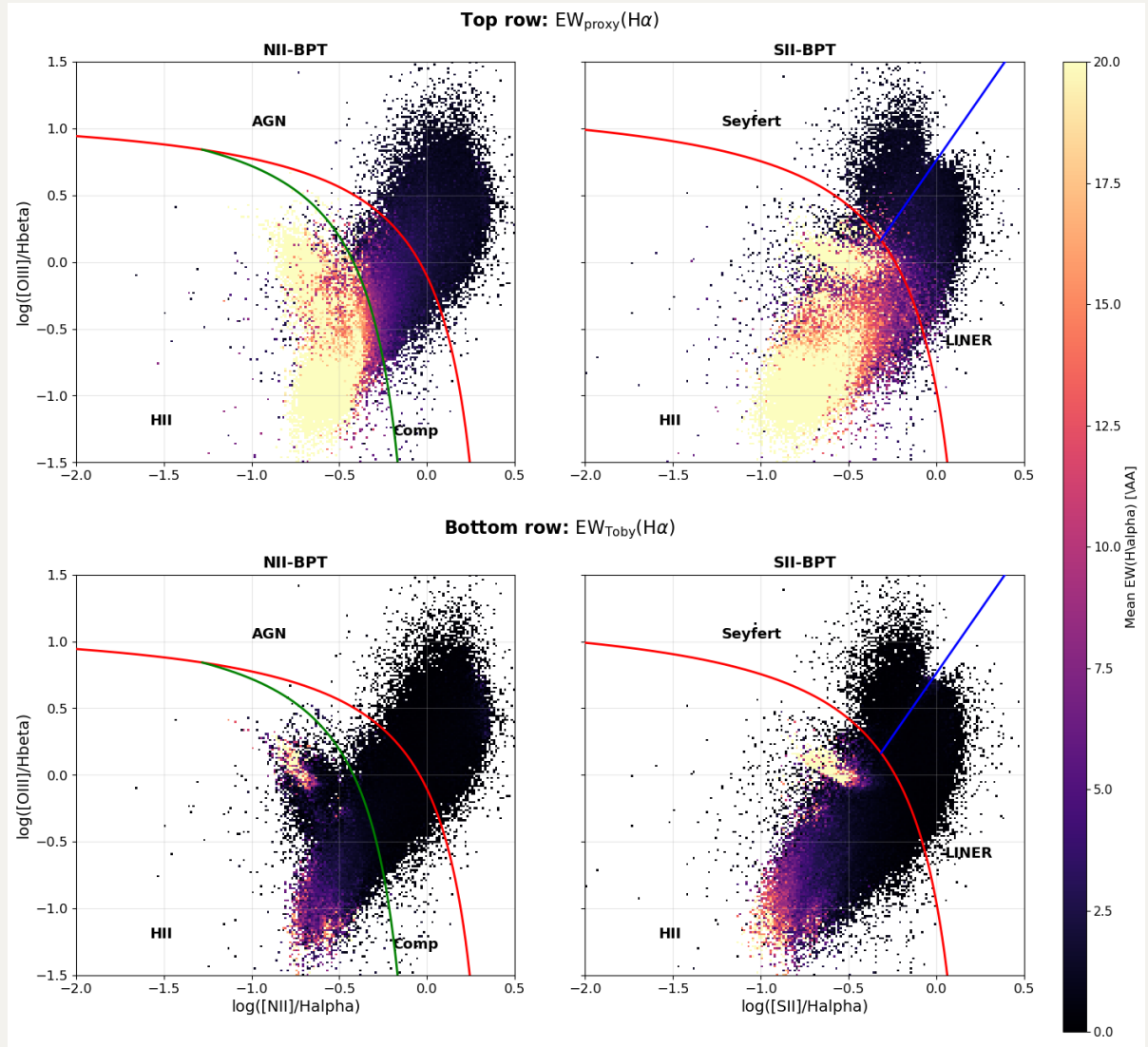
2. Comparison with $\text{EW}_{\text{pseudo}}(\text{H}\alpha)$

So the visual check looks fine. Both $\text{EW}_{\text{proxy}}(\text{H}\alpha)$ and $\text{EW}_{\text{pseudo}}(\text{H}\alpha)$ are very identical.



3. Comaprison with Toby's EW(H α)





4. Plots in paper with new $EW_{\text{proxy}}(\text{H}\alpha)$

